From Confusion to Confidence: One Hour of Simulation, Weeks of Growth

Accelerating Neuraxial Skills with Targeted Training

Mass General Brigham

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The Issue → The Innovation

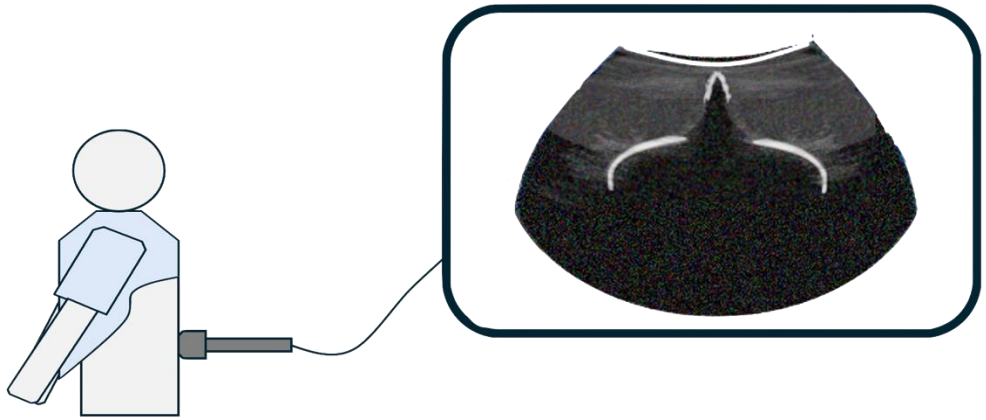
The Issue: Variability

Inconsistent Exposure

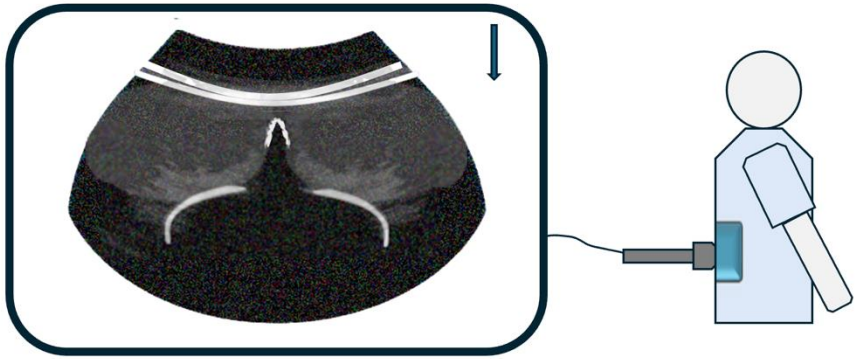
Case volume and difficulty vary significantly. Returning rotators assigned ‘hard’ placements.

Practice Utility Limited

Residents easy to palpate and ultrasound, patient population more challenging



The Innovation: Simulation Jacket



Targeted Deliberate Practice

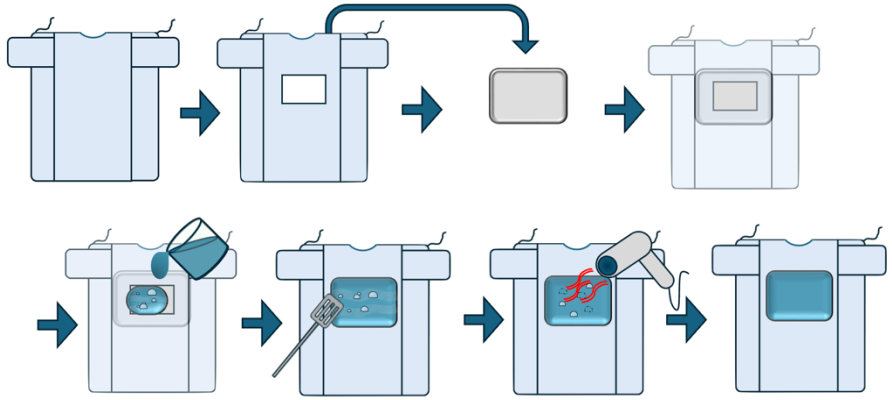
Realistically mimics challenges without requiring coordination with outside parties

Standardizes Early Experience

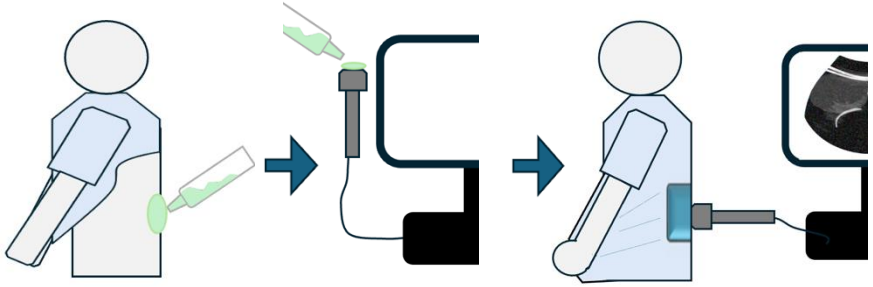
Does not depend on availability of patients or favor giving ‘difficult’ cases to senior learners

Adjustable Depth With Real, Live Anatomy

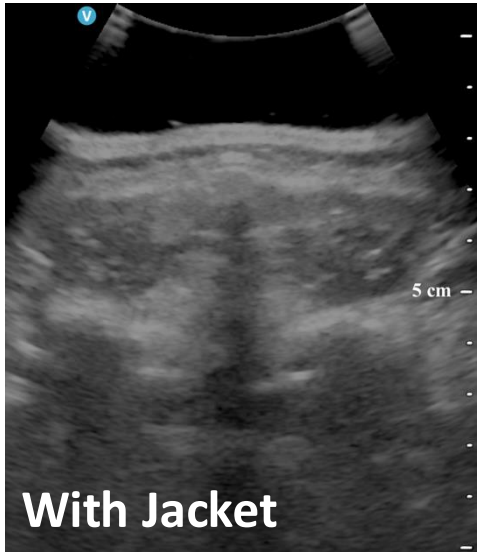
Building A Sim Jacket



Using A Sim Jacket

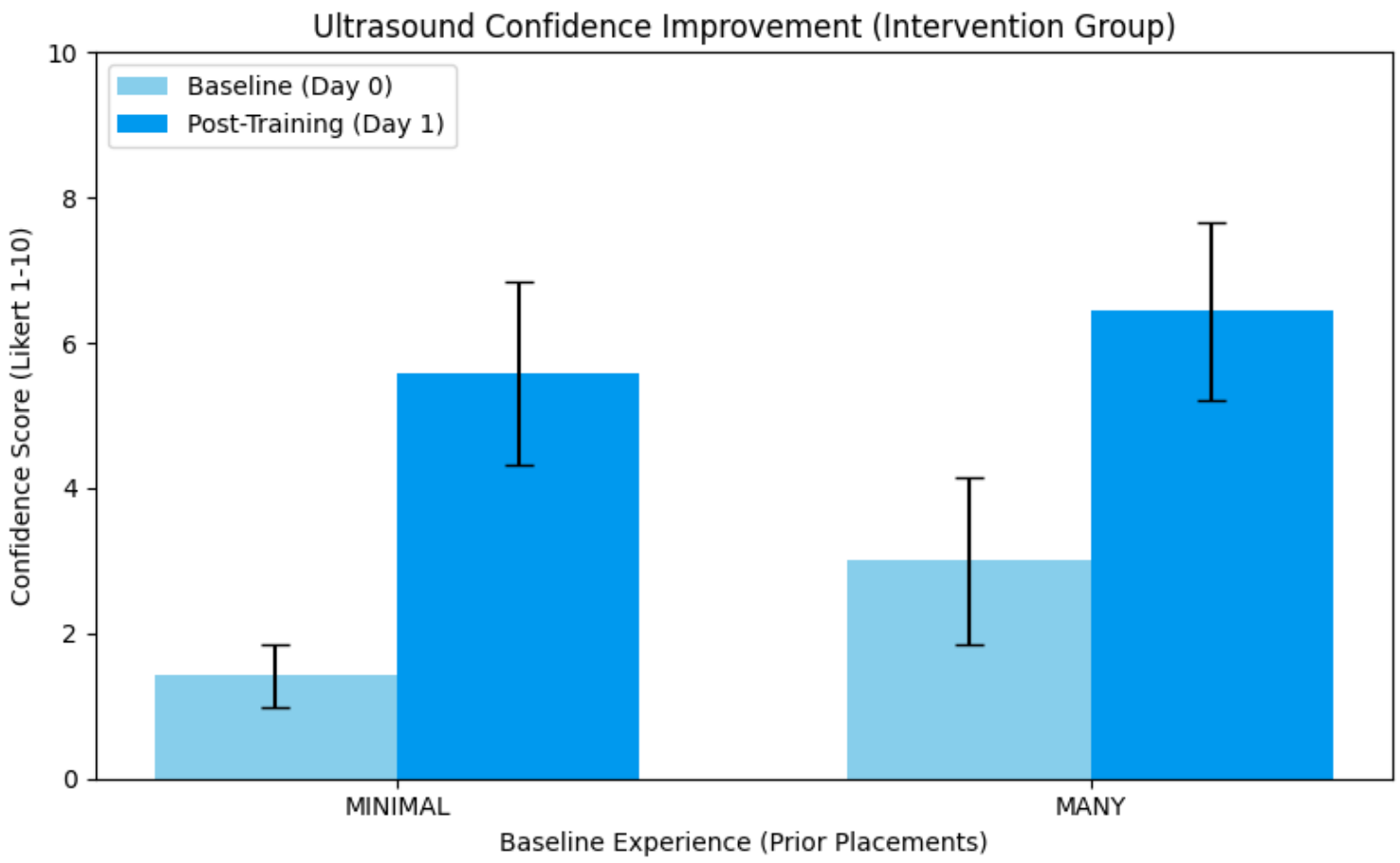


Real Images - Same Person:



- Simulates challenging palpation and ultrasound
- Titratable depth for progressive difficulty
- Enables rapid-cycle deliberate practice
- Easy to replicate with over-the-counter tools

Results: All Trainee Levels Build US Skills



Large Effect Size

d = 1.25

Cohen's d
(Intervention vs Baseline)

Statistically Significant

p = 0.005

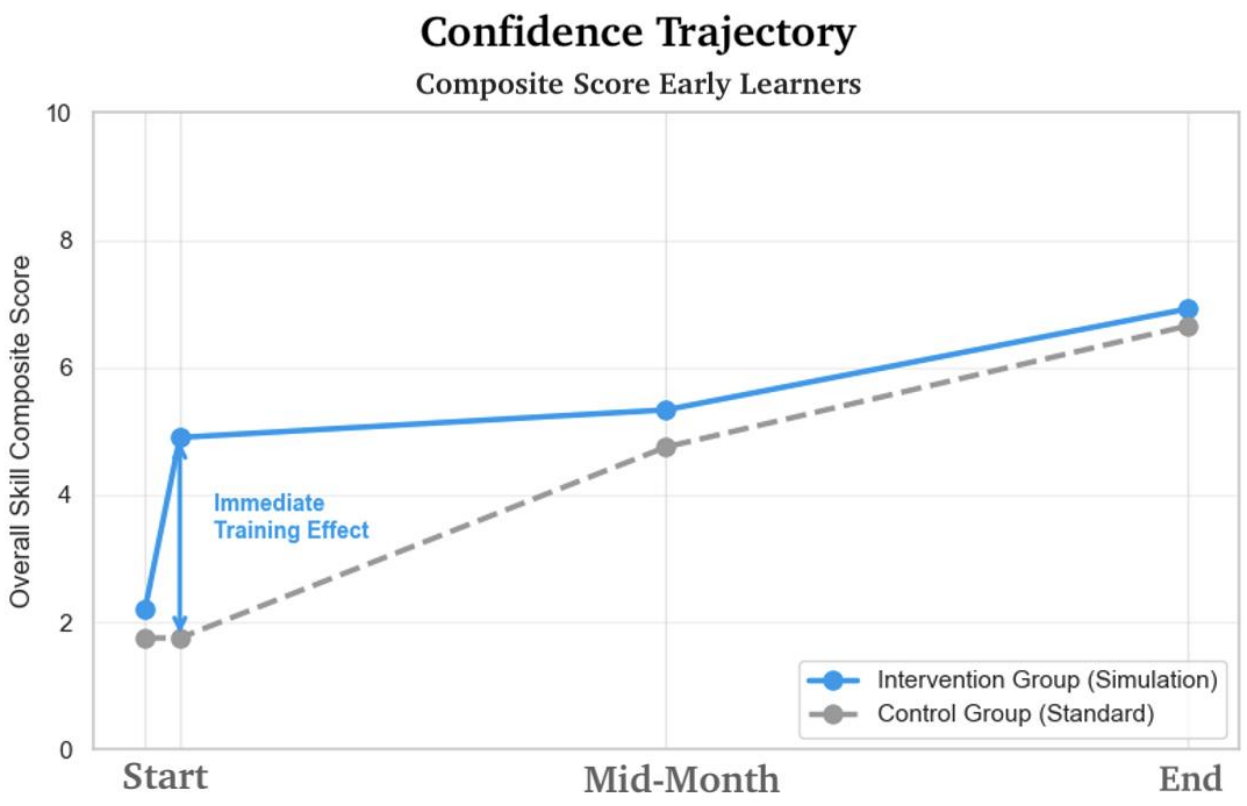
All Experience Groups
(Intervention vs Baseline)

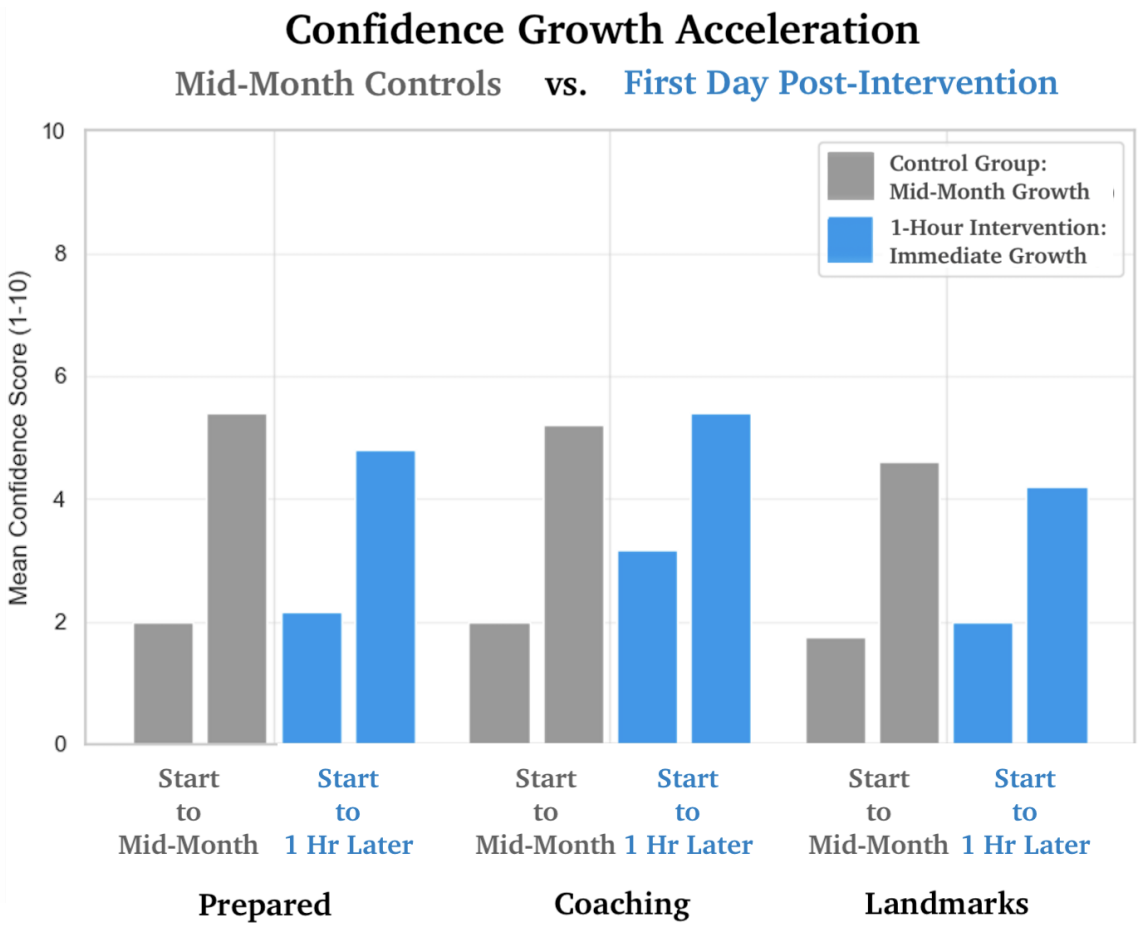
Confidence Boost

23% → **88%**

Reporting Confidence ≥ 5
(Intervention vs Baseline)

Fast-Track Novices: Comparable to 2 Weeks

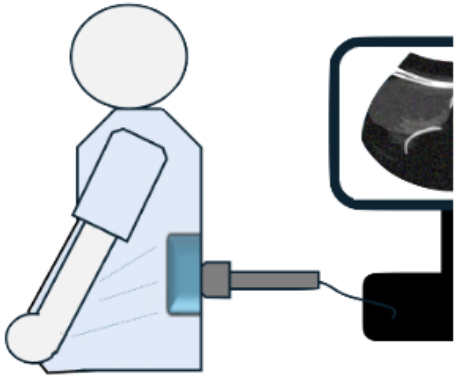




Impact, Persistence, and Invitation – Try It!

Enduring Skills

- **Accelerates Early Learning:** trained group gains similar to two-weeks of standard training
- **Diversity of Experience, Normalized Exposure:** multiple exams, varying difficulty for everyone
- **Skills Persist:** immediate and sustained gains

Easy to Implement and Scale

- **Low Resource Load:** jacket made with low-cost over-the-counter tools and consumables, all other equipment is standard on L&D (gel, ultrasound, learners, attendings)
- **Broad Application:** can be used as adjunct to static training models, adaptable to non-OB
- **Efficient:** 1-hour session, immediate results